

TreeScore

Measuring India's urban green cover — clearly, honestly, and consistently. A civic-data product that starts with three Mumbai localities and a simple discipline: measure first, claim later.

| *"Let's measure first."*

Prepared for	Stage	Date
Abhinandan Tejaswi	Real-data beta · v0.2 boundaries	September 2026

01 What TreeScore Is

TreeScore is a mobile-first civic-data platform that turns satellite imagery into an honest, comparable measure of urban green cover — starting in Mumbai, one locality at a time.

The founding insight came from watching Mumbai turn into a concrete jungle: the coastal road, shrinking mangroves, metro works, and roughly 40 trees lost daily to construction. The original idea was to map the exact number of trees around any location and nudge citizens to plant more. Through our analysis that idea was sharpened into something more defensible and more useful.

What it is

- A transparent measurement layer for urban green cover
- A suburb-vs-suburb comparison and ranking engine
- A foundation for citizen, government and corporate action

What it is not

- Not an activism platform
- Not an official government map
- Not a claim of legal or administrative boundaries
- Not (yet) exact tree-canopy or tree-count data

THE CORE DISCIPLINE

Credibility comes from restraint. Every boundary is labelled a "TreeScore-defined analysis area, not an official administrative boundary," and every figure is a "satellite-derived green cover estimate" — never presented as an exact tree count. This honesty is the product's moat, not a legal footnote.

02 The Strategic Picture

Three hats — entrepreneur, VC, and change agent — produced one consistent verdict: real demand, viable B2B monetisation, and behaviour change only if an action layer sits on top of the map.

Why the timing is unusually good

The BMC has floated a tender for Mumbai's first tree census in eight years, expected to begin in early 2026 and run roughly 18 months, recording canopy spread, height, species, health and geo-tagged coordinates. Government data portals are historically hard to use, and the BMC was criticised for never releasing the 2018 census publicly. That creates an opening: **TreeScore doesn't need to be the census — it needs to be the interface and action layer on top of it.** Launching as the census rolls out means every news cycle about it markets the product.

Demand — born of anger, which is the durable kind

Mumbai's green cover sits below 13%, with per-capita open space around 1.1 sq m against the WHO norm of 9. Tree plantations fell about 72% year-on-year in 2024-25. The Save Aarey movement proved citizens will mobilise at scale. AQI apps went from niche to ubiquitous in five years for the same reason a "Green Score" can: a single number that quantifies environmental anxiety and invites comparison.

HONEST CAVEAT

Demand to *look* is proven; demand to *act repeatedly* is not. Every prior Indian tree-mapping effort plateaued at hobbyist scale because it asked citizens to do data entry. TreeScore inverts this: satellite and government data do the work; citizens consume and act.

How it makes money

Revenue line	Who pays & why	Signal
CSR verification (anchor line)	Corporates must spend 2% of profits on CSR; the biggest failure mode is the plantation drive nobody monitors. TreeScore's geo-tagged survival tracking is the fix.	Indian CSR hit a record ₹34,909 cr in FY24; environment is a top category.
Municipal / govt contracts	A canopy-loss monitoring layer between censuses, sold to BMC / MMRDA / other ULBs.	BMC budget ₹74,427 cr; only ~112 tree officers for 25+ lakh trees.
Societies & developers (later)	RWAs paying for green audits; developers paying for "green locality scores" in listings.	Walkability scores exist; green scores don't.
Citizens	Free. They are distribution and data-pressure, never the revenue.	Indian consumer LTV is low; civic apps monetise the data layer.

Realistic scale: ₹50 lakh–1 crore ARR in 18–24 months via CSR verification, government contracts as a slower second line. A sustainable mission-driven company (₹5–10 cr/yr range), more impact-investor / climate-angel fundable than classic venture-scale. Be honest about that shape.

Does it actually drive tree planting?

Only if the app removes friction rather than merely raising awareness. The behavioural chain that works:

Outrage → Identity → Low-friction action → Visible status → Repeat. The map supplies outrage ("you lost 340 trees"), the ranking supplies identity and competition (the Swachh-rankings playbook — no corporator wants to rank last), and a one-tap layer ("a drive 800 m away, Saturday, 47 joined — RSVP") converts attention into trees. Tracking *survival*, not just planting, is the honest high ground nobody else holds.

03 Key Product Decisions Locked

Several forks were resolved during the work. These are the settled defaults the build now rests on.

Decision	Resolution	Reasoning
Interface	App/web primary, WhatsApp as the engagement layer	Maps and rankings are visual and exploratory; WhatsApp is for transactions (RSVP, reminders, photo upload). Sequence, don't choose.
MVP data depth	Level 1 (canopy %) ships first; Level 2 (individual trees) built in parallel	Canopy % via Sentinel-2 NDVI is the global standard and can't be disputed on methodology. Tree counts are attackable at launch — and collide with the BMC census.
Metric framing	"Satellite-derived green cover estimate," not "tree canopy"	NDVI captures trees, shrubs, lawns, grass and mangroves alike. Naming it honestly protects credibility.
First localities	Andheri West Core, Powai Lake + Urban Area, Juhu	Small, honest beta. Founder's home turf. Expand suburb-by-suburb, letting rankings drive competition.
Boundaries	TreeScore-defined analysis areas, never official	Legal safety and methodological truth — locality edges are genuinely subjective.

CANOPY IS THE PRODUCT METRIC; TREES ARE THE BUSINESS METRIC

Citizens and media rank suburbs by canopy %. Corporates and the BMC pay for tree-level verification. Both are needed eventually — canopy simply ships first because it's fast and unarguable.

04 What Has Been Built So Far

The first real-data foundation: validated locality boundary files for three Mumbai localities, ready to feed Google Earth Engine, plus a starter NDVI analysis script and full documentation.

DELIVERABLE · V0.1

mumbai-localities-v0.1.geojson

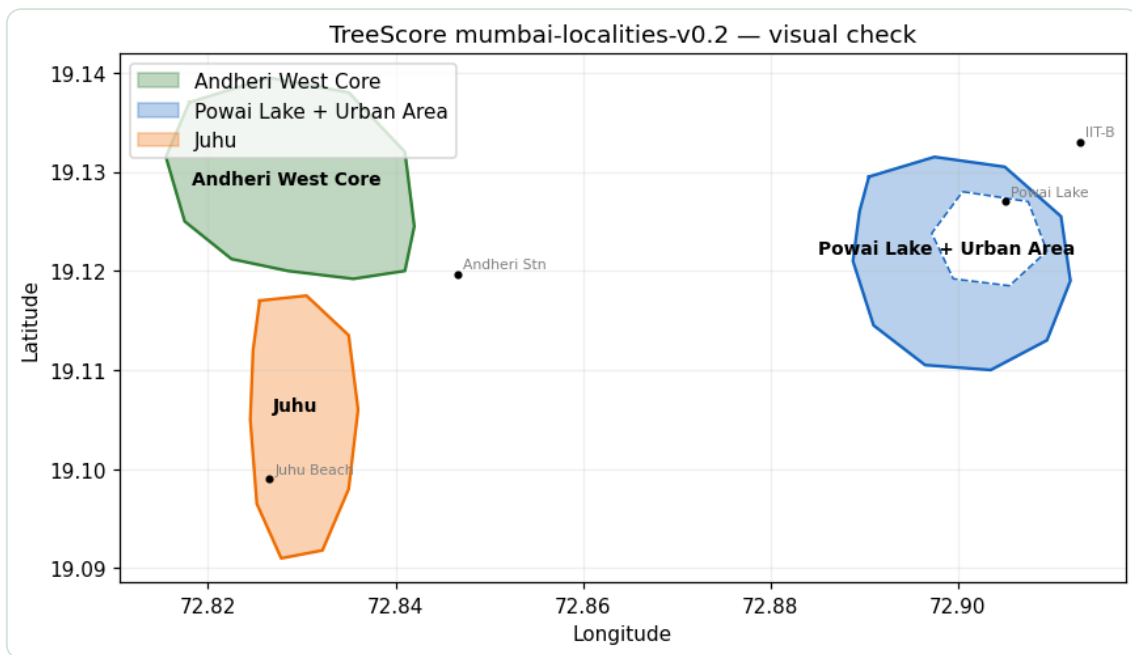
Three approximate polygons (Andheri West, Powai, Juhu) as a valid WGS84 FeatureCollection with honesty metadata on every feature. Validated for JSON/GeoJSON correctness, coordinate order, ring closure, self-intersection, and — after catching and fixing a 2.25 sq km Andheri–Juhu overlap — zero inter-locality overlap. Shipped with a README and a Google Earth Engine canopy script.

DELIVERABLE · V0.2 (CURRENT)

mumbai-localities-v0.2.geojson

A refined, more honest version driven by what the v0.1 inspection revealed:

- **Andheri West** → **Andheri West Core**. Tightened to the true western core (Seven/Four Bungalows, DN Nagar, Azad Nagar, station-side west), rather than implying it covers all of Andheri West including Lokhandwala/Oshiwara.
- **Powai** → **Powai Lake + Urban Area**. Pulled back from the IIT-Bombay interior and forest-edge green that had inflated v0.1's reading, focused on the Hiranandani / lake-edge / JVLR urban fabric, with the lake cut out as an inner ring and Vikhroli / SGNP / Aarey excluded.
- **Juhu**. Kept conservative; beach edge nudged east so less sand and sea sit inside the land denominator.
- New honesty fields added throughout: `parentLocality`, `measurementLabel`, `publicDisclaimer`. Language shifted from "tree canopy" to "green cover estimate."



v0.2 analysis areas with landmark reference points. Andheri West Core sits west of Andheri Station; Powai pulls back from the IIT-B interior with the lake cut out; Juhu stays small and coastal.

Validated area summary (v0.2)

Analysis area	localityId	Area (approx)	vs v0.1
Andheri West Core	andheri-west-core	4.81 sq km	tightened from 5.46
Powai Lake + Urban Area	powai-lake-urban-area	3.57 sq km (land, lake excluded)	tightened from 6.17
Juhu	juhu	2.83 sq km	tightened from 3.46

All three pass every validation check: valid geometry, closed rings, correct lon/lat order, no self-intersection, no mutual overlap, and the Powai lake hole confirmed inside its outer ring.

The full file set

```
data/boundaries/  
├─ mumbai-localities-v0.1.geojson # first three polygons  
├─ mumbai-localities-v0.2.geojson # refined, current  
├─ README.md # v0.1 documentation  
├─ README-v0.2.md # v0.2 documentation  
├─ boundary-assumptions-v0.2.md # per-locality reasoning  
├─ treescore-canopy-v0.1.gee.js # Earth Engine NDVI script  
├─ preview.png # v0.1 visual check  
└─ preview-v0.2.png # v0.2 visual check
```

HONEST STATUS OF THE BOUNDARIES

All three areas are **medium confidence**: correctly placed in coordinate space, but hand-reasoned approximations, not basemap-traced lines. They are good enough to run a first NDVI pass — not good enough to publish a number without visual inspection in geojson.io or QGIS first, threshold calibration, and water masking.

05 Roadmap — What Needs to Be Done

Sequenced so that each step ships something usable and defensible, and so the product is the established civic interface by the time BMC census data lands.

PHASE 0

Foundation & ground truth

IN PROGRESS

- **Visually inspect v0.2** in geojson.io / QGIS over a satellite basemap; confirm each polygon and the Powai lake hole.
- **File the RTI** to BMC's K-West ward for the 2018 tree-census data and tree-felling permissions 2019–2025 (₹10; 30-day legal window). Likely both a dataset and a launch headline.
- **Talk to 15–20 people** — NGO leaders (Vanashakti, Aarey groups), housing-society secretaries, BMC/forest contacts, regular residents — before more building.

PHASE 1

Calibrated Level-1 canopy MVP

NEXT · 4–6 WKS

- **Threshold calibration helper** — test NDVI thresholds against a known park vs. a known market inside the polygons, so the first figure rests on a calibrated threshold, not a default 0.35.
- **Run the GEE pipeline** — dry-season Sentinel-2 median composite, water masking, per-locality green-cover % with CSV export.
- **Ship the web app** — drop a pin, see the green-cover estimate for your locality, plus the three-way comparison. Mobile-responsive PWA, not native.
- **Build the "green cover lost since 2015" view** — Sentinel-2 archive is free back to 2015; the loss story is the shareable, press-worthy hook.

PHASE 2

Rankings & expansion

3–4 MONTHS

- Add suburbs in order of civic activism and media attention: Bandra West → Juhu-extended → Versova → Powai-extended, then respond to requests.
- Tighten priority boundaries from *medium* to *high* confidence (basemap-traced) so cross-locality rankings become publishable.
- Launch the locality ranking board — the competition engine that drives repeat engagement and political pressure.
- Begin the Level-2 individual-tree detection layer quietly (DeepForest / Meta-WRI model + ground-truth walks) for a second press cycle.

PHASE 3**Action layer**

4–6 MONTHS

- Partner with 2–3 NGOs already running drives; TreeScore becomes their volunteer sign-up and tracking tool (they get free infrastructure, you get users and credibility).
- Add the WhatsApp engagement layer: drive RSVPs, reminders, geo-tagged photo uploads.
- Add civic-action features: report illegal felling, track BMC replantation promises, ward-level green rankings for corporators/MLAs.
- Introduce "adopt a mapped tree/pit" with quarterly photo updates.

PHASE 4**Monetisation**

6–12 MONTHS

- Build the CSR verification dashboard — geo-tagged plantation tracking with quarterly survival rates (the recurring-revenue product).
- Approach 3–5 Mumbai corporates with active CSR (Godrej, TCS, JSW, Mahindra) offering monitored, verified plantation programs (~₹3–10 lakh/yr each).
- Open the municipal monitoring conversation with BMC / MMRDA once the census data is public and TreeScore is the established interface.
- Ingest BMC census data and merge with satellite monitoring — becoming the canonical "official count + what's changed since" layer.

06 Immediate Next Actions

If only three things happen next, these are they — ordered.

#	Action	Why now
1	Inspect v0.2 boundaries in geojson.io / QGIS and note any edges to retrace	Nothing downstream is trustworthy until the polygons are eyeballed on a real basemap.
2	File the K-West RTI (census data + felling permissions)	Runs in the background on a 30-day clock; feeds both the dataset and the launch story.
3	Build the NDVI threshold-calibration helper, then run the first GEE canopy pass	Converts the boundary files into the first real, defensible green-cover numbers.

ONE HONEST EXPECTATION TO SET

When the calibrated pipeline runs, tightened Powai will likely read *lower* than v0.1 — that is the intended effect of removing forest-edge green, and it is the more truthful number. TreeScore's job is to report the number that's true, not the one that flatters.

Disclaimer. All locality boundaries referenced in this document are TreeScore-defined analysis areas, not official administrative boundaries, and do not correspond to BMC ward limits or any legal demarcation. All green-cover figures are satellite-derived estimates (an NDVI-based proxy that captures all vegetation, not only trees) and must not be described as exact tree-canopy area or tree counts. Market, CSR, budget and policy figures cited are drawn from the research conducted during this project and should be re-verified before external use. Boundary areas are approximate, computed from hand-reasoned polygons at medium confidence.